

ABCDs of Nutritional Evaluation

Part B: Biomarkers and Functional Labs



ELIZABETH BOHAM, MD, RD



MICHAEL STONE, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Elizabeth Boham, MD, RD, disclosed no relevant financial relationships with any ineligible companies.

Michael Stone, MD, is a consultant for Tailored Script and Calroy Health.

Unless otherwise noted, all stock photos in this presentation are used with permission by 123RF.com.

Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Cite conventional laboratory biomarkers that provide evidence of nutritional status and patient risk profile.
2. Explain the challenges inherent in the interpretation of reference ranges for selected micronutrients and other biomarkers of nutritional status.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Interpret laboratory results of selected biomarkers of nutritional status.

Clinical Focus

- Laboratory evaluation of nutrient imbalance
- CBC and CMP



ABCDs of the Functional Nutrition Evaluation



Anthropometrics
& Vital Signs



**Biomarkers &
Functional Labs**

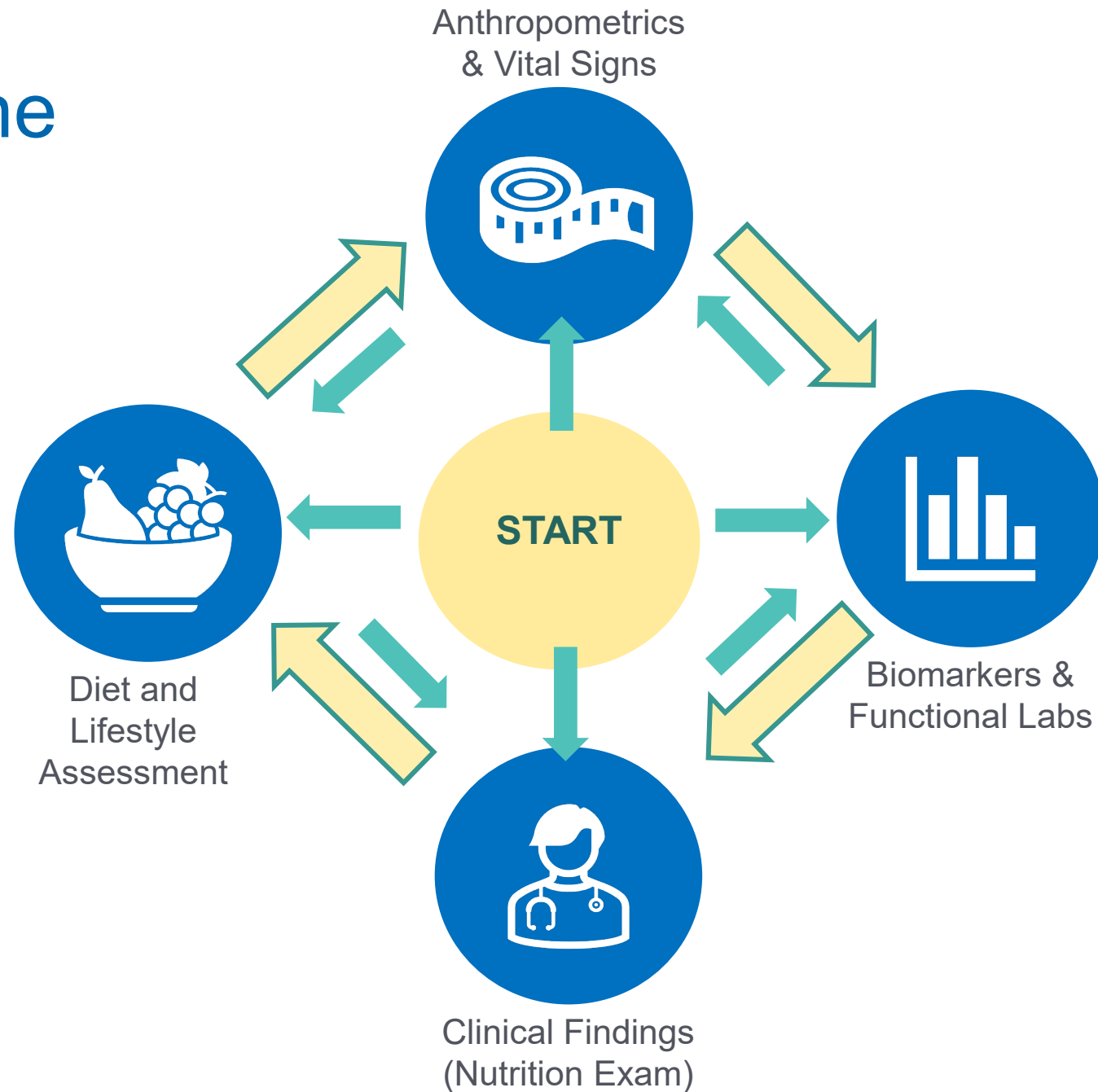


Clinical Findings
(Nutrition Exam)

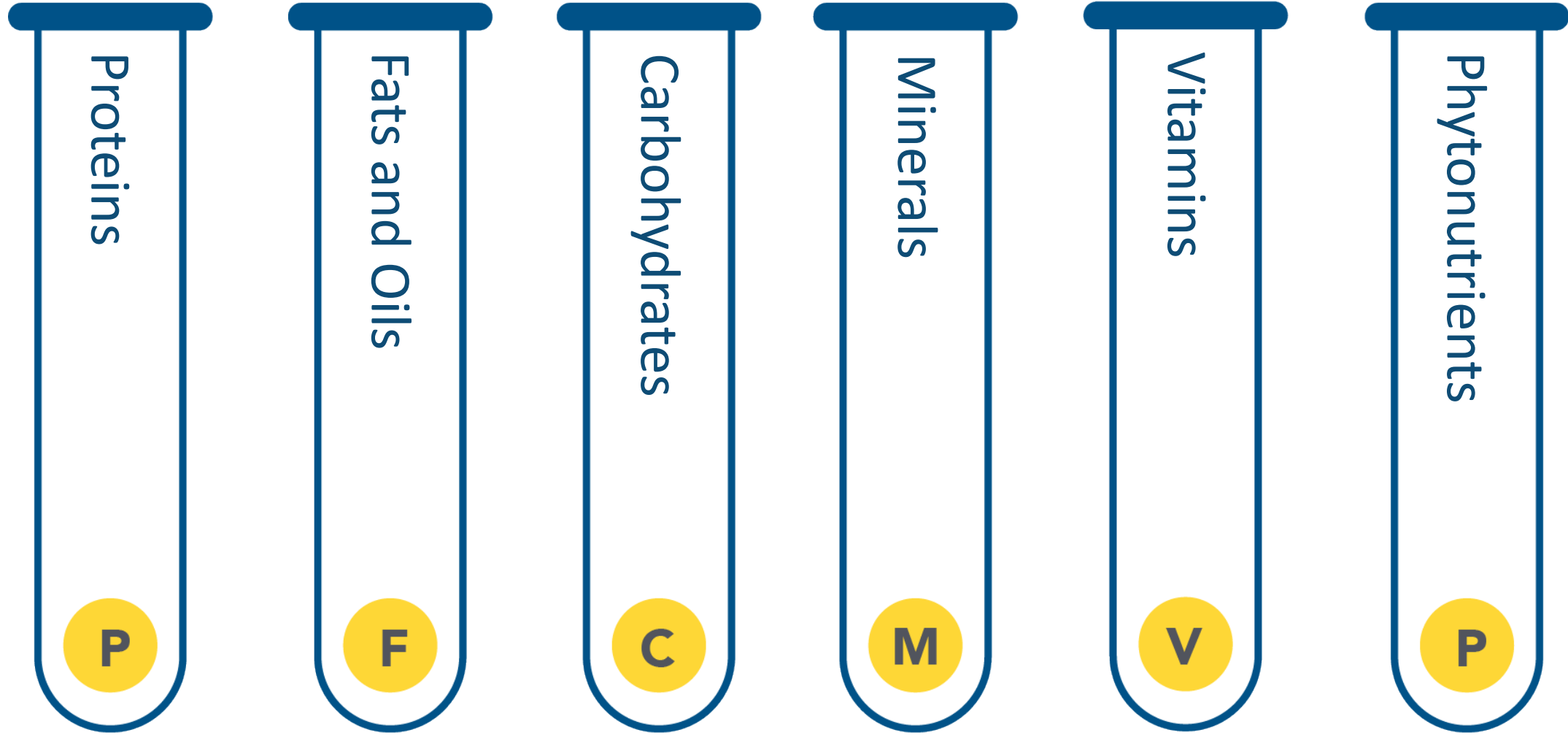


Diet and Lifestyle
Assessment

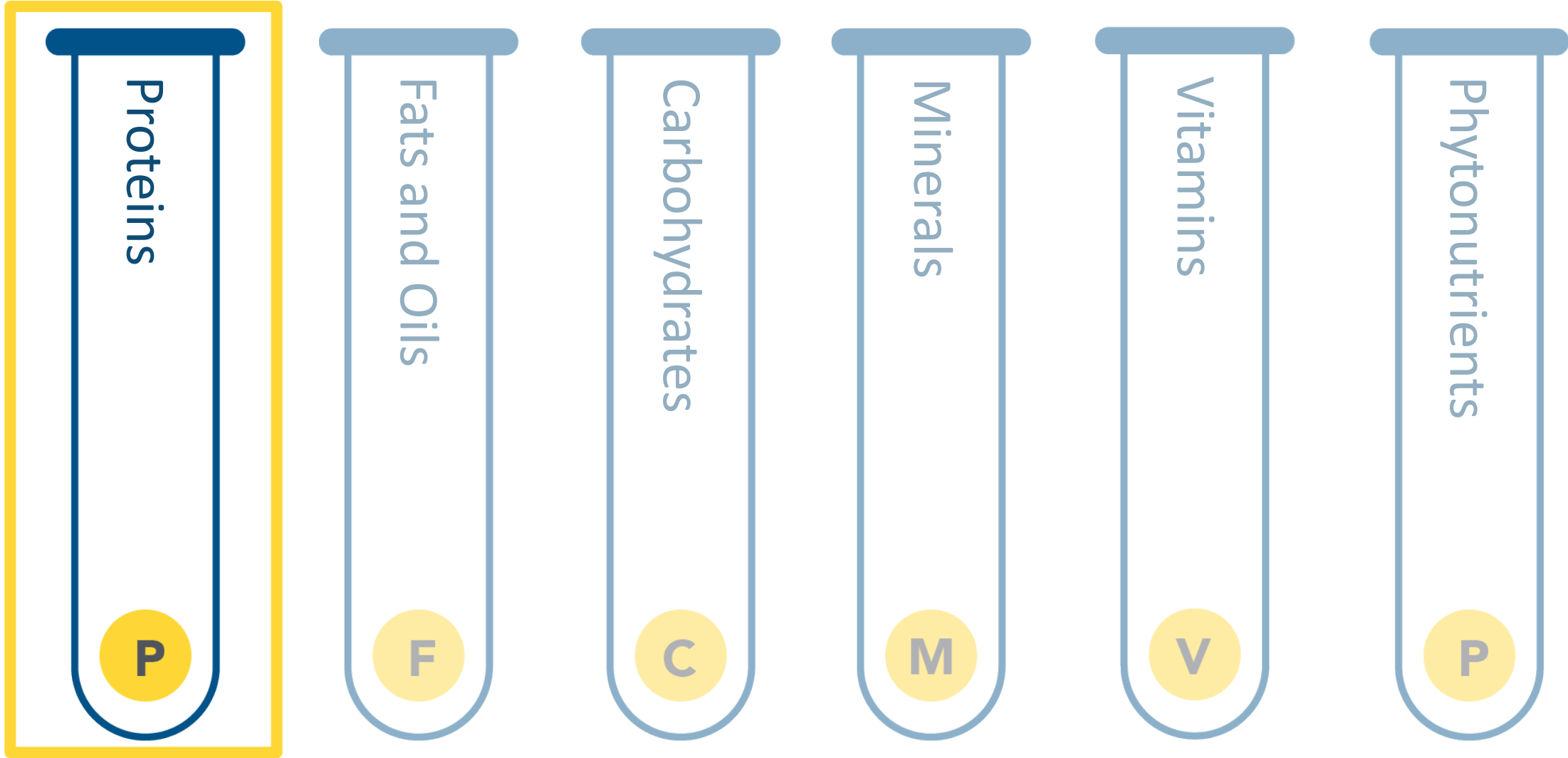
ABCDs of the Functional Nutrition Evaluation



BIOMARKERS AND LABORATORY ASSESSMENT



BIOMARKERS AND LABORATORY ASSESSMENT



Laboratory Biomarkers

Protein

- Total protein
- Albumin
- Globulin

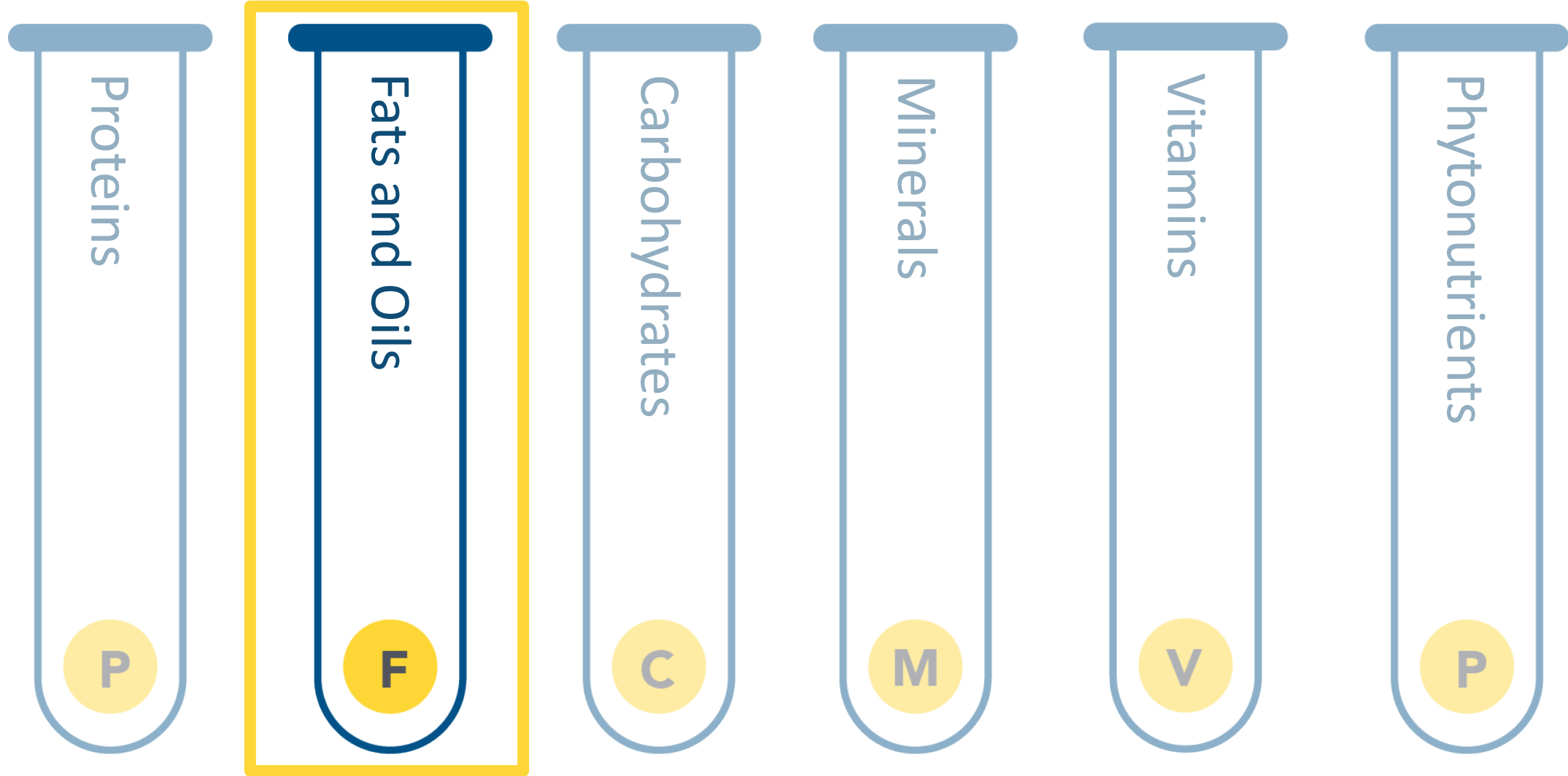


Clinical Focus

- Laboratory evaluation of nutrient imbalance
- Lipid profile



BIOMARKERS AND LABORATORY ASSESSMENT

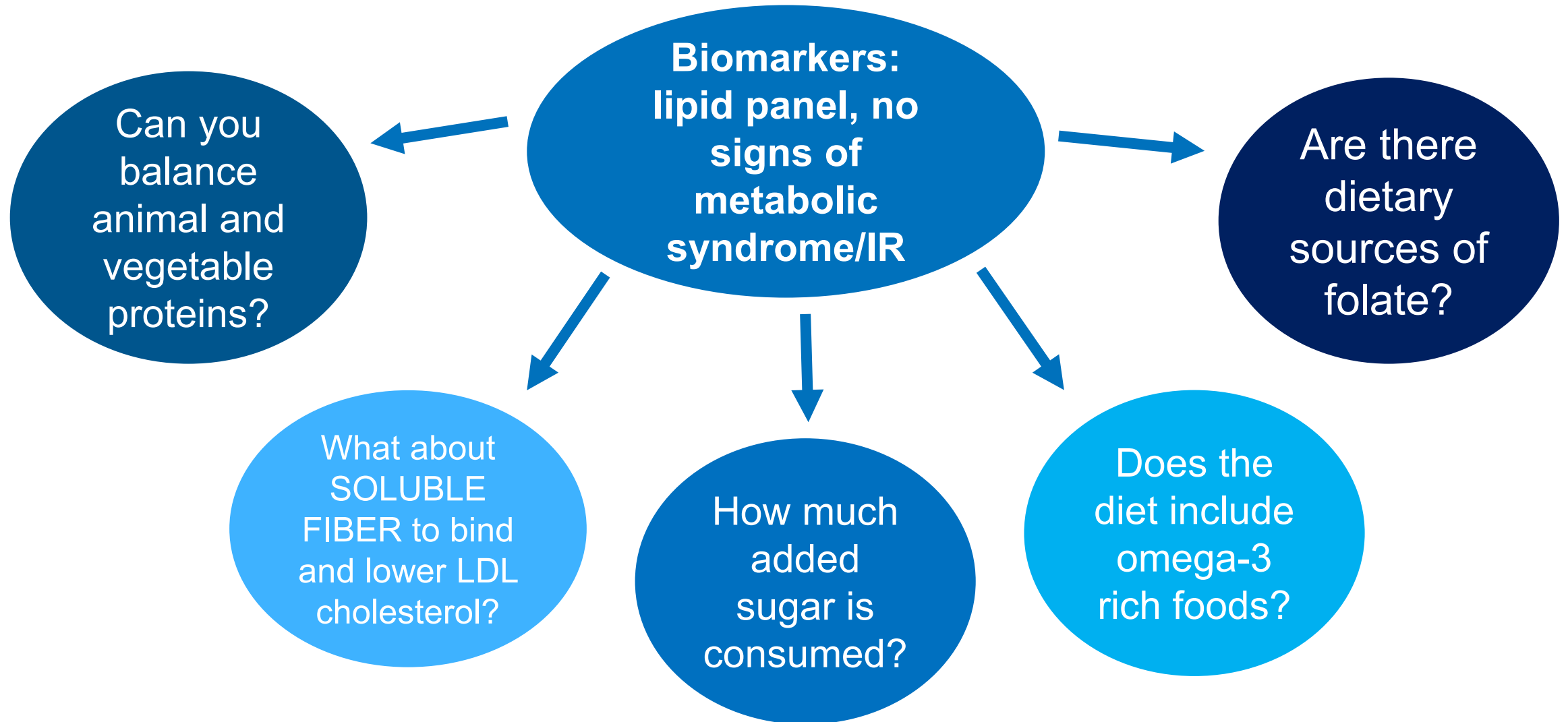


Laboratory Biomarkers

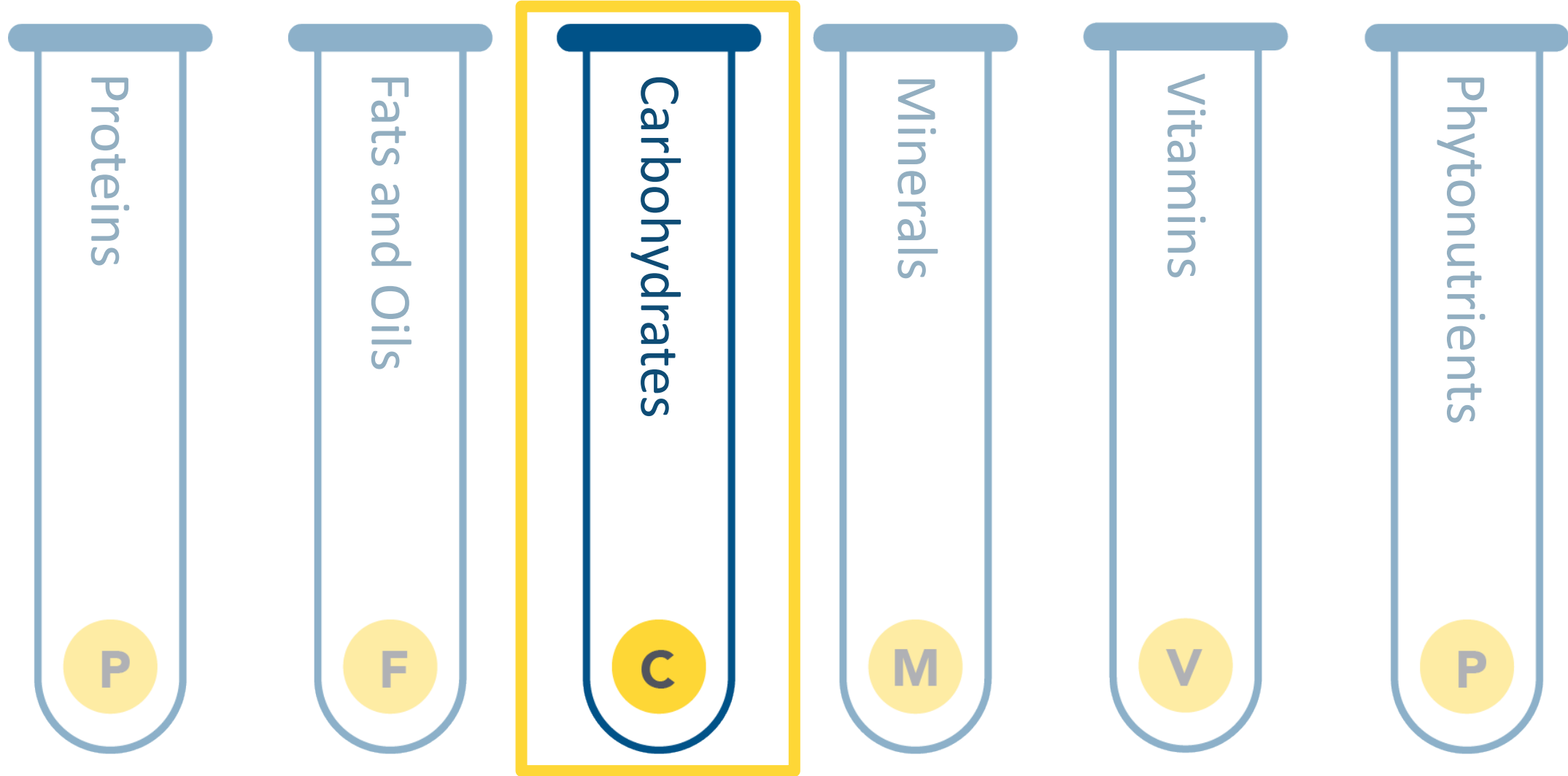
Fats

- **Lipid profile and ratios:**
 - Total cholesterol (TC)
 - Low density lipoprotein (LDL)
 - High density lipoprotein (HDL)
 - Triglycerides (TG)
- TG/HDL ratio
- Lipid particle number, size, and subfractions
- Apolipoprotein B (ApoB)
- Apolipoprotein A1 (ApoA1)
- ApoB/A1 ratio

Hyperlipidemia



BIOMARKERS AND LABORATORY ASSESSMENT

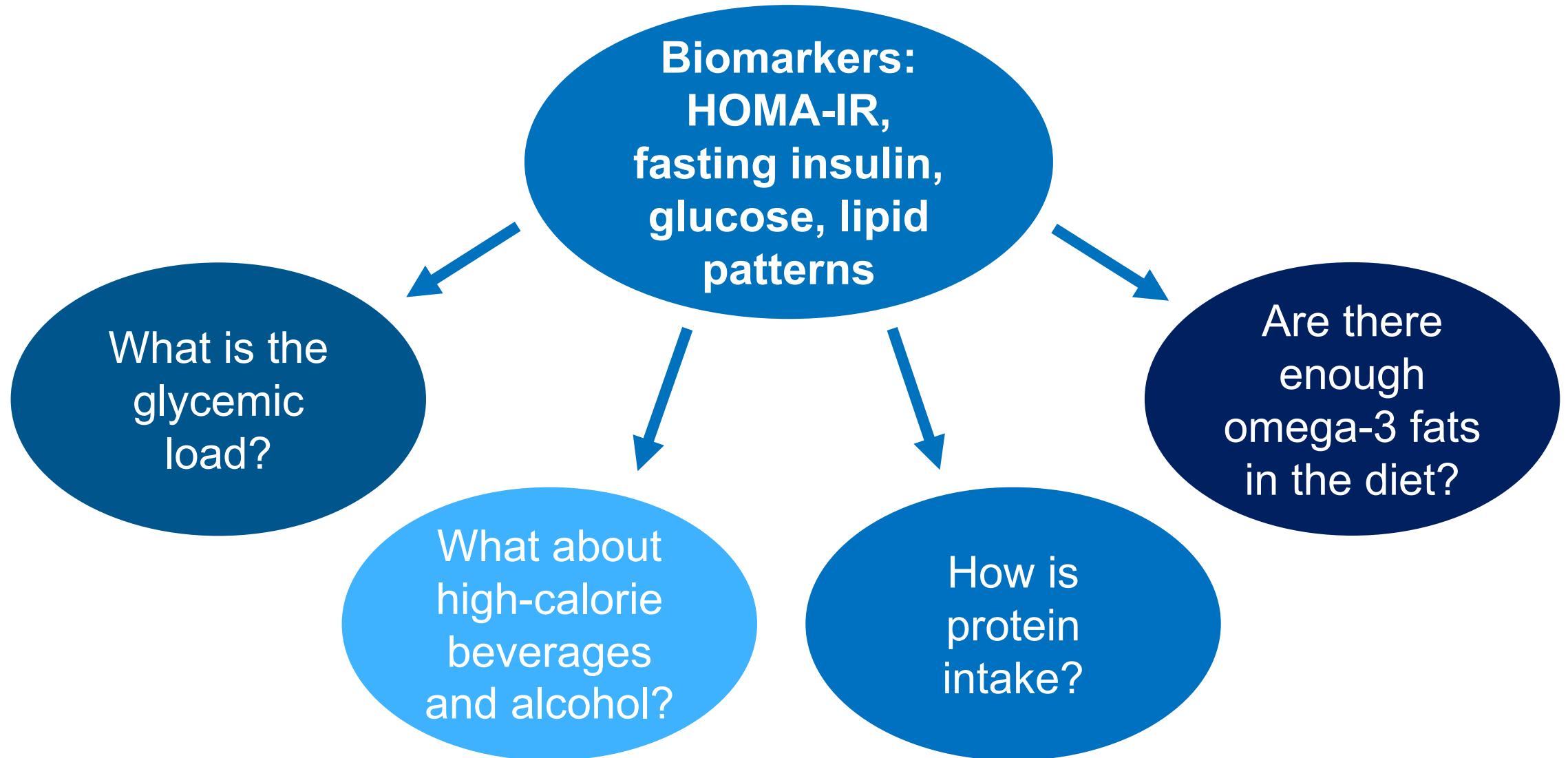


Laboratory Biomarkers

Carbohydrates

- Fasting blood glucose
- Fasting insulin
 - Homeostatic model assessment of insulin resistance (HOMA-IR) score
- Hemoglobin A1c (HgA1c)
- Triglycerides
- Gamma-glutamyl transferase (GGT)
- Uric acid

Metabolic Syndrome



Laboratory Tests

Protein – Fats - Carbohydrates

Laboratory Tests	
Protein	• Total protein • Albumin • Globulin
Fats	• Lipid profile and ratios: • Total cholesterol (TC) • Low density lipoprotein (LDL) • High density lipoprotein (HDL) • Triglycerides (TG) • Lipid particle number, size, and subfractions • TG/HDL ratio • Apolipoprotein B (ApoB) • Apolipoprotein A1 (ApoA1) • ApoB/A1 ratio
Carbohydrates	• Fasting blood glucose • Hemoglobin A1c (HgA1c) • Triglycerides • Gamma-glutamyl transferase (GGT) • Uric acid

ABCDs of Nutritional Evaluation

Part B: Biomarkers and Functional Labs

Part2

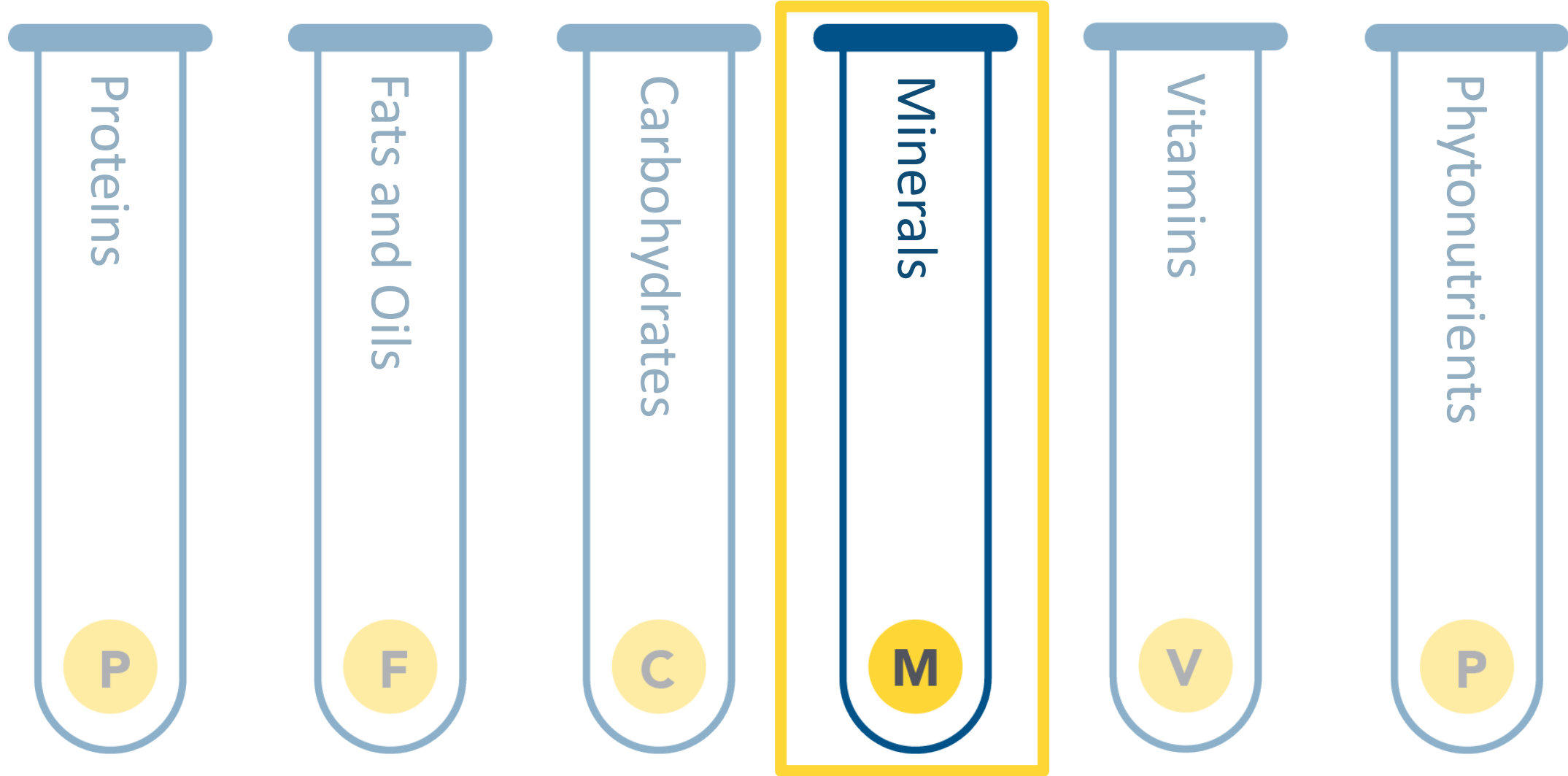
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Laboratory evaluation of nutrient imbalance
- Selected minerals, vitamins, and phytonutrients



BIOMARKERS AND LABORATORY ASSESSMENT



Laboratory Biomarkers

Minerals

- Complete blood count with differential (CBC with differential)
- Comprehensive metabolic panel (CMP)
- Calcium
 - Serum calcium (Ca^{2+})
 - Ionized calcium
- Copper
 - Serum copper
 - Ceruloplasmin

Laboratory Biomarkers

Minerals

- Iron
 - Serum iron
 - Transferrin
 - Total iron-binding capacity (TIBC)
 - Ferritin
 - Complete blood count (CBC)
 - Hemoglobin (Hgb)
 - Hematocrit (Hct)
 - Mean corpuscular volume (MCV)
 - Mean corpuscular hemoglobin (MCH)
 - Hemochromatosis (HFE) gene panel

Laboratory Biomarkers

Minerals

- Magnesium
 - Serum magnesium (Mg^{2+})
 - Red blood cell (RBC) magnesium
- Phosphorus
 - Serum phosphorus
- Selenium
 - Serum selenium
- Zinc
 - Plasma zinc (Zn^{2+})

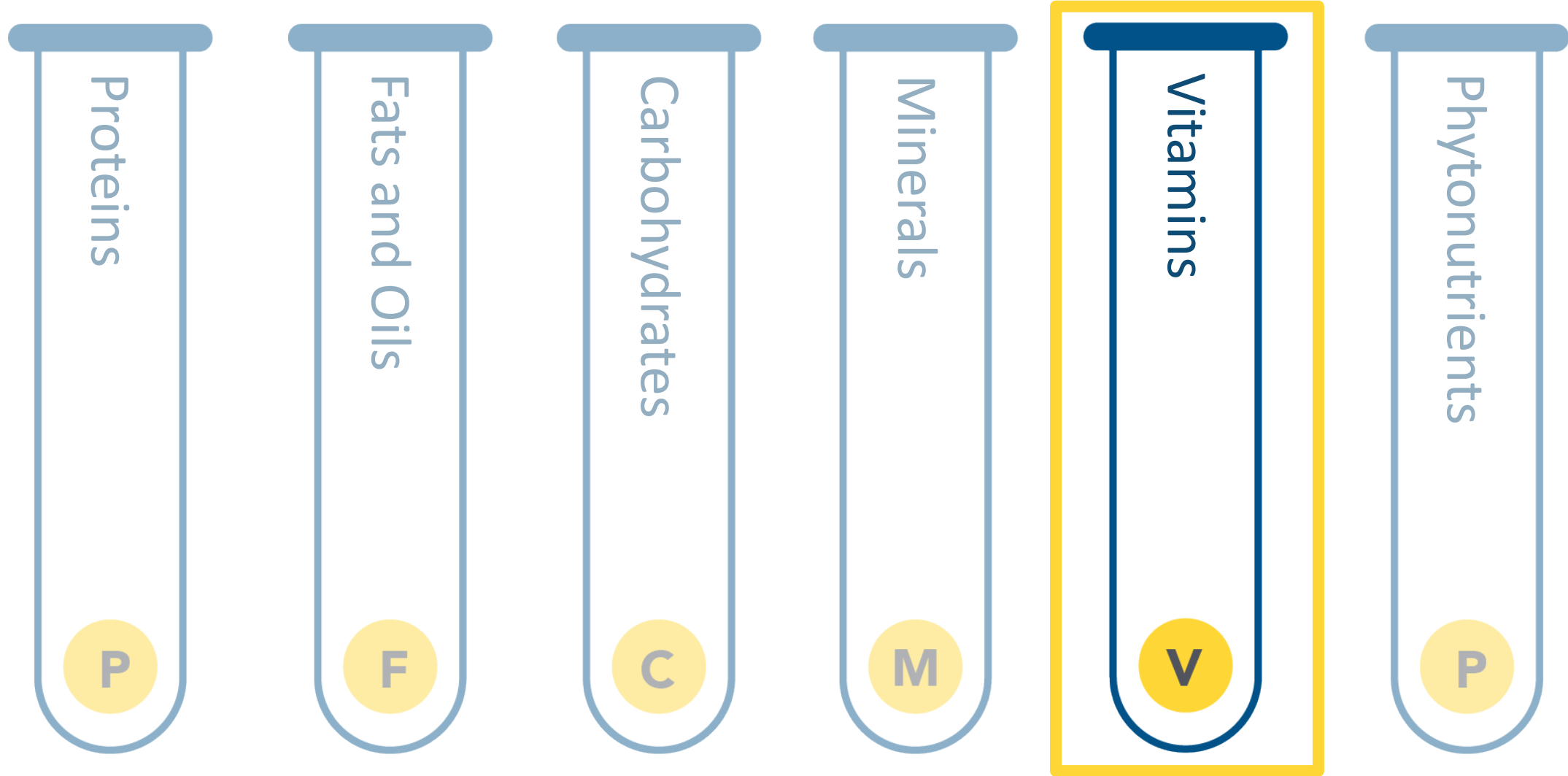
Laboratory Tests Minerals

Minerals	Laboratory Tests
General	• Complete blood count with differential (CBC w/differential) • Comprehensive metabolic panel (CMP)
Calcium	• Serum calcium (Ca²⁺) • Ionized calcium
Copper	• Serum copper • Ceruloplasmin
Iron	• Serum iron • Transferrin • Total iron-binding capacity (TIBC) • Ferritin • Complete blood count (CBC) • Hemoglobin (Hgb) • Hematocrit (Hct) • Mean corpuscular volume (MCV) • Mean corpuscular hemoglobin (MCH) • Hemochromatosis (HFE) gene panel

Laboratory Tests Minerals

Minerals	Laboratory Tests
Magnesium	• Serum magnesium (Mg^{2+}) • Red blood cell (RBC) magnesium
Phosphorus	• Serum phosphorus
Selenium	• Serum selenium
Zinc	• Plasma zinc (Zn^{2+})

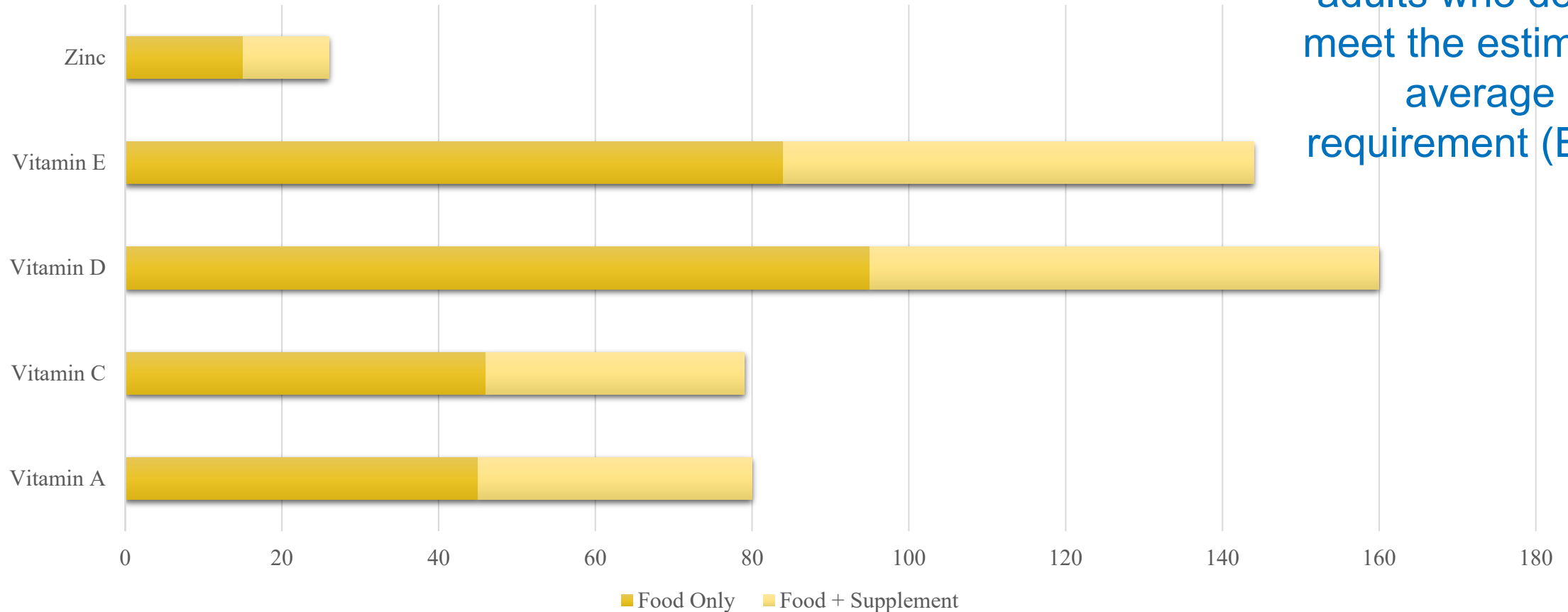
BIOMARKERS AND LABORATORY ASSESSMENT



Prevalence of Vitamin Deficiencies

2005-2016 NHANES Adult Nutrient Inadequacies

Percentage of US adults who do not meet the estimated average requirement (EAR)



Long Description: Prevalence of Vitamin Deficiencies

- A horizontal stacked bar graph. The horizontal axis represents estimated average requirement. The vertical axis includes the nutrients Zinc, and vitamins E, D, C, and A. The key takeaway is that a significant percentage of U S adults do not meet the estimated average requirement for these nutrients.

Laboratory Biomarkers

Fat-Soluble Vitamins

- Complete blood count (CBC) with differential
- Vitamin D
 - Serum 25-hydroxy vitamin D3 [25-(OH)D3]
- Vitamin K
 - Prothrombin time (PT)
 - Partial thromboplastin time (PTT)

Laboratory Biomarkers

Water-Soluble Vitamins

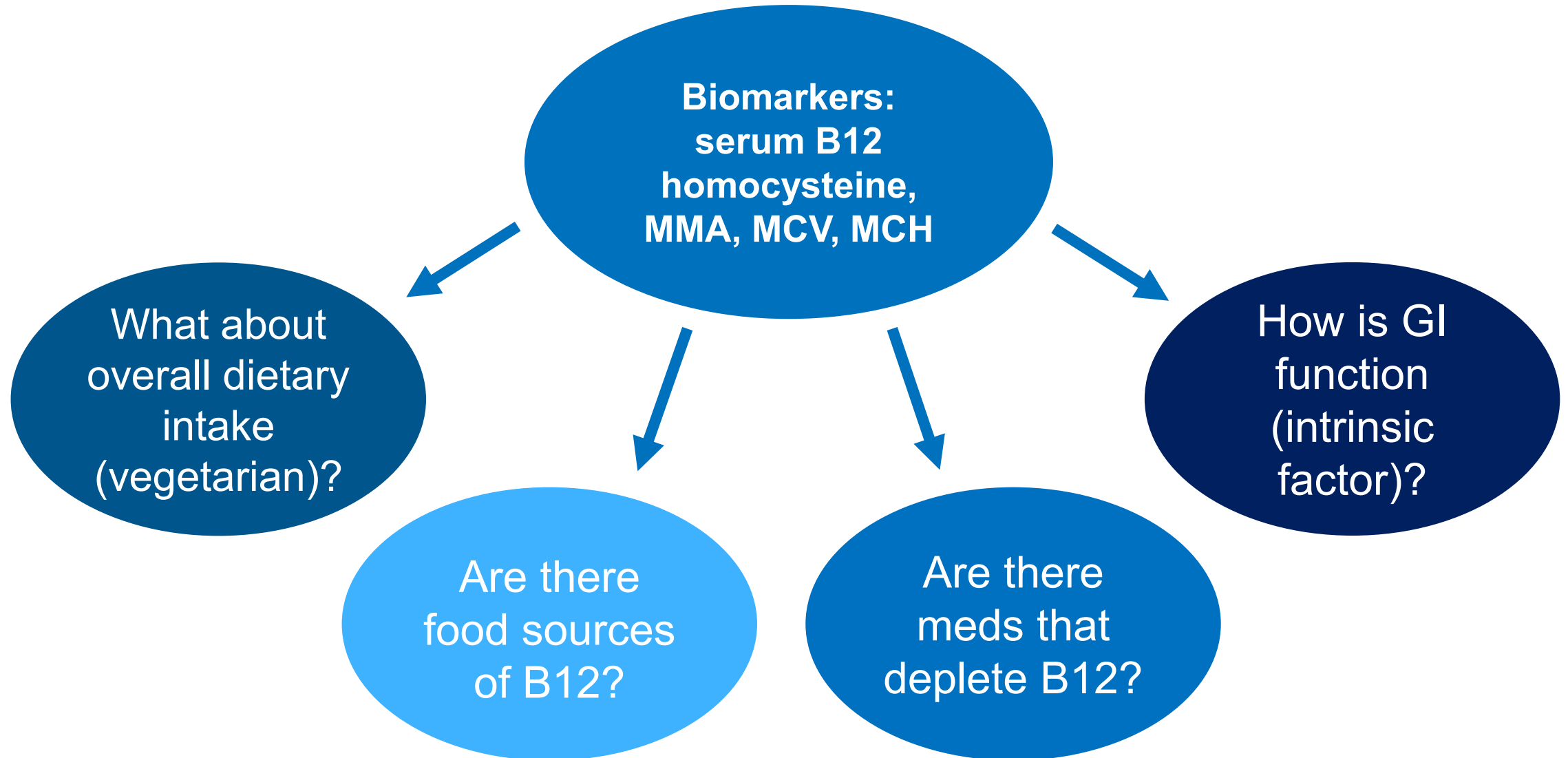
- Vitamin B1 (Thiamine)
 - Plasma thiamine
- Vitamin B2 (Riboflavin)
 - Plasma riboflavin
- Vitamin B6 (Pyridoxine)
 - Homocysteine
 - Plasma pyridoxal 5-phosphate (P5P)

Laboratory Biomarkers

Water-Soluble Vitamins

- Vitamin B9 (Folate)
 - Serum folate
 - Homocysteine
 - Mean corpuscular volume (MCV)
- Vitamin B12 (Cobalamin)
 - Serum B12
 - Homocysteine
 - Methylmalonic acid (MMA)
 - Mean corpuscular volume (MCV)
 - Mean corpuscular hemoglobin (MCH)

Vitamin B12 Deficiency

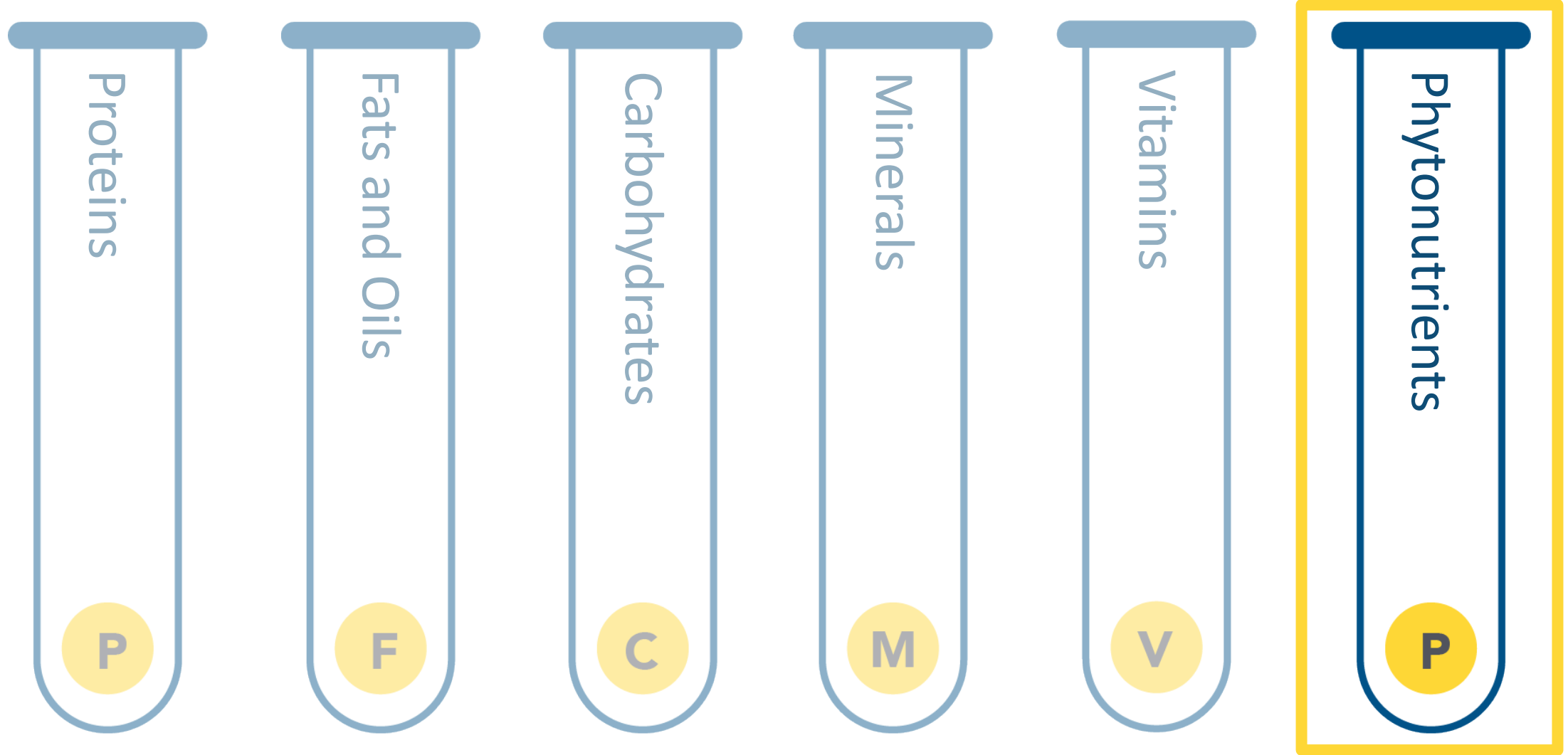


Laboratory Tests

Vitamins

Vitamins	Laboratory Tests
General	Complete blood count with differential (CBC w/differential)
Vitamin D	Serum 25-hydroxy vitamin D3 [25-(OH)D3]
Vitamin K	- Prothrombin time (PT) - Partial thromboplastin time (PTT)
Vitamin B1 (Thiamine)	Plasma thiamine
Vitamin B2 (Riboflavin)	Plasma riboflavin
Vitamin B6 (Pyridoxine)	- Homocysteine - Plasma pyridoxal 5-phosphate (P5P)
Vitamin B9 (Folate)	- Serum folate - Homocysteine - Mean corpuscular volume (MCV)
Vitamin B12 (Cobalamin)	- Serum B12 - Homocysteine - Methylmalonic acid (MMA) - Mean corpuscular volume (MCV) - Mean corpuscular hemoglobin (MCH)

BIOMARKERS AND LABORATORY ASSESSMENT



Discretionary Tests in Complex Cases

Phytonutrients

Polyphenols

- Urinary metabolites:
 - Daidzein
 - Genistein
 - Glycitein
 - Hydroxytyrosol

Carotenoid

- Plasma levels of carotenoids:
 - Beta-carotene
 - Lycopene
 - Lutein

Oxidized Low-Density Lipoprotein (OxLDL)

- Marker of oxidative stress that decrease with polyphenol intake

C-Reactive Protein (CRP)

- A marker of inflammation that decreases with polyphenol intake

Discretionary Tests in Complex Cases

Phytonutrients

Sulfated and Glucuronidated Metabolites

- Glucuronidated daidzein and sulfated compounds of:
 - Pyrogallol
 - Ferulic Acid
 - Enterolactone

Catecholamines

- Levels of catecholamines, particularly dopamine, increase with phytonutrient intake

Specific Flavonoid Metabolites

- Epigallocatechin gallate intake:
 - 1,3,5-trimethoxybenzene
 - 4-O-methylgallic acid
 - 3-O-methylgallic acid
 - Gallic acid
- Quercetin intake:
 - Urinary 4-ethylphenol
 - Benzoic acid
 - 4-ethylbenzoic acid

References

PHYTONUTRIENTS: DISCRETIONARY TESTS IN COMPLEX CASES

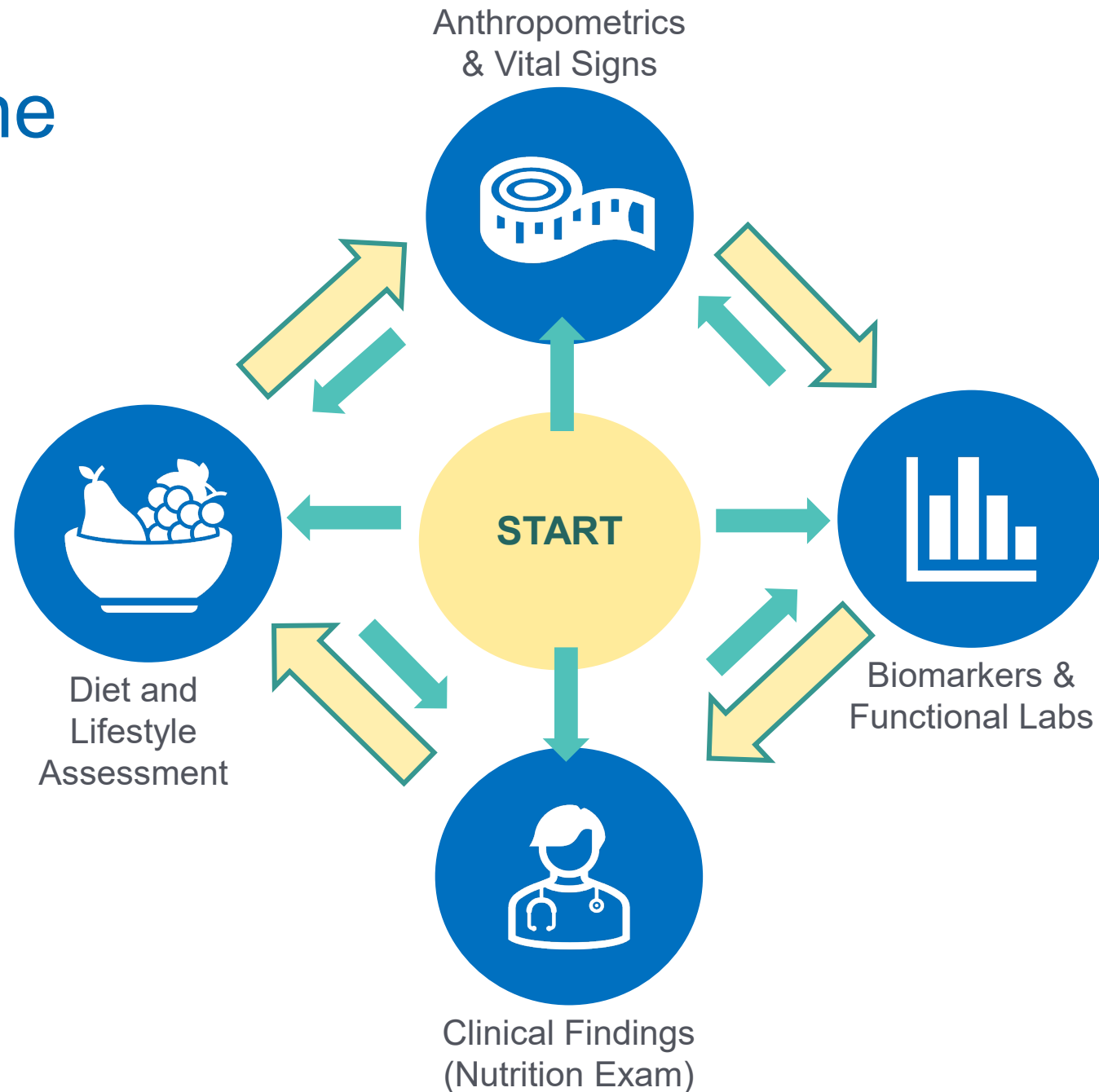
- Arcusa R, Carrillo JÁ, Xandri-Martínez R, et al. Effects of a fruit and vegetable-based nutraceutical on biomarkers of inflammation and oxidative status in the plasma of a healthy population: a placebo-controlled, double-blind, and randomized clinical trial. *Molecules*. 2021;26(12):3604. doi:10.3390/molecules26123604 (funded by the Juice Plus Company)
- Burri BJ, Neidlinger TR, Clifford AJ. Serum carotenoid depletion follows first-order kinetics in healthy adult women fed naturally low carotenoid diets. *J Nutr*. 2001;13(8):2096-2100. doi:10.1093/jn/131.8.2096
- Loke WM, Jenner AM, Proudfoot JM, et al. A metabolite profiling approach to identify biomarkers of flavonoid intake in humans. *J Nutr*. 2009;139(12):2309-2314. doi:10.3945/jn.109.113613
- Pérez-Jiménez J, Hubert J, Hooper L, et al. Urinary metabolites as biomarkers of polyphenol intake in humans: a systematic review. *Am J Clin Nutr*. 2010;92(4):801-809. doi:10.3945/ajcn.2010.29924
- Tsiara I, Hervás Povo B, Alotaibi W, Young Tie Yang P, Rodriguez-Mateos A, Globisch D. Characterizing the sulfated and glucuronidated (poly)phenol metabolome for dietary biomarker discovery. *J Agric Food Chem*. 2025;73(11):6702-6710. doi:10.1021/acs.jafc.4c12596
- Woodside JV, Draper J, Lloyd A, McKinley MC. Use of biomarkers to assess fruit and vegetable intake. *Proc Nutr Soc*. 2017;76(3):308-315. doi:10.1017/S0029665117000325

Clinical Focus

- Conclusion, summary, and key takeaways.



ABCDs of the Functional Nutrition Evaluation



Key Clinical Takeaways



Conventional laboratory test results from the complete blood count, the chemistry panel, and the lipid profile can yield insights into the nutritional status of a patient.



Laboratory reference ranges to detect frank nutrient deficiencies may differ from optimal reference ranges that detect nutrient insufficiencies capable of mediating physiological dysfunctions.



Laboratory testing for specific nutrients should be judiciously initiated when indicated to inform clinical decision making.

Laboratory Tests

Protein – Fats - Carbohydrates

Laboratory Tests	
Protein	• Total protein • Albumin • Globulin
Fats	• Lipid profile and ratios: • Total cholesterol (TC) • Low density lipoprotein (LDL) • High density lipoprotein (HDL) • Triglycerides (TG) • Lipid particle number, size, and subfractions • TG/HDL ratio • Apolipoprotein B (ApoB) • Apolipoprotein A1 (ApoA1) • ApoB/A1 ratio
Carbohydrates	• Fasting blood glucose • Hemoglobin A1c (HgA1c) • Triglycerides • Gamma-glutamyl transferase (GGT) • Uric acid

Laboratory Tests Minerals

Minerals	Laboratory Tests
General	• Complete blood count with differential (CBC with differential) • Comprehensive metabolic panel (CMP)
Calcium	• Serum calcium (Ca²⁺) • Ionized calcium
Copper	• Serum copper • Ceruloplasmin
Iron	• Serum iron • Transferrin • Total iron-binding capacity (TIBC) • Ferritin • Complete blood count (CBC) • Hemoglobin (Hgb) • Hematocrit (Hct) • Mean corpuscular volume (MCV) • Mean corpuscular hemoglobin (MCH) • Hemochromatosis (HFE) gene panel

Laboratory Tests Minerals

Minerals	Laboratory Tests
Magnesium	• Serum magnesium (Mg^{2+}) • Red blood cell (RBC) magnesium
Phosphorus	• Serum phosphorus
Selenium	• Serum selenium
Zinc	• Plasma zinc (Zn^{2+})

Laboratory Tests

Vitamins

Vitamins	Laboratory Tests
General	Complete blood count with differential (CBC with differential)
Vitamin D	Serum 25-hydroxy vitamin D3 [25-(OH)D3]
Vitamin K	- Serum copper - Ceruloplasmin
Vitamin B1 (Thiamine)	Plasma thiamine
Vitamin B2 (Riboflavin)	Plasma riboflavin
Vitamin B6 (Pyridoxine)	- Homocysteine - Plasma pyridoxal 5-phosphate (P5P)
Vitamin B9 (Folate)	- Serum folate - Homocysteine - Mean corpuscular volume (MCV)
Vitamin B12 (Cobalamin)	- Serum B12 - Homocysteine - Methylmalonic acid (MMA) - Mean corpuscular volume (MCV) - Mean corpuscular hemoglobin (MCH)

ABCDs of Nutrition Evaluation

Part B: Biomarkers and Functional Labs

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE